Data Collection and Preprocessing Phase

Section	**Description**
Data Overview	The dataset contains restaurant information from Zomato, including name, reviews, ratings, cuisines, cost, and more. The data is cleaned, deduplicated, and preprocessed for building a content-based recommendation system.
Resizing	*Not applicable for text data.*
Normalization	Ratings are normalized to a 1-5 scale using MinMaxScaler. Text is lowercased and punctuation is removed.
Data Augmentation	Not applicable for text data.
Denoising	Text is cleaned by removing newline characters and punctuation.
Edge Detection	Not applicable for text data.

Date	4 April 2026
Name	Karuna Uttam PAtil
Project Title	Restaurant Recommendation System

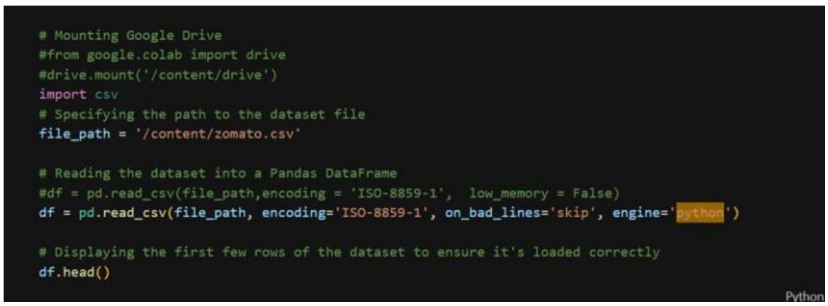
Maximum Marks	6 Marks

Data Preprocessing

The images will be preprocessed by resizing, normalizing, augmenting, denoising, adjusting contrast, detecting edges, converting color space, cropping, batch normalizing, and whitening data. These steps will enhance data quality, promote model generalization, and improve convergence during neural network training, ensuring robust and efficient performance across various computer vision tasks.

Color Space Conversion	Not applicable for text data.
Image Cropping	Not applicable for text data.
Batch Normalization	Not applicable for text data.

Data Preprocessing Code Sc reenshots

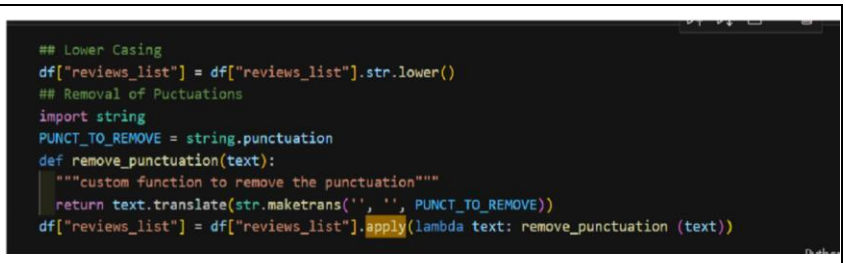
Loading Data	

```python
# Mounting Google Drive
#from google.colab import drive
#drive.mount('/content/drive')
import csv
# Specifying the path to the dataset file
file_path = '/content/zomato.csv'

# Reading the dataset into a Pandas DataFrame
#df = pd.read_csv(file_path,encoding = 'ISO-8859-1',  low_memory = False)
df = pd.read_csv(file_path, encoding='ISO-8859-1', on_bad_lines='skip', engine='python')

# Displaying the first few rows of the dataset to ensure it's loaded correctly
df.head()
```

Resizing	*Not applicable*
Normalization	

```
# Computing Mean Rating
restaurants = list(df['name'].unique())
df['Mean Rating'] = 0
for i in range(len(restaurants)):
  df['Mean Rating'][df['name'] == restaurants[i]] = df['rate'][df['name'] == restaurants[i]].mean()
#Scaling the mean rating values
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler (feature_range = (1,5))
df[['Mean Rating']] = scaler.fit_transform(df[['Mean Rating']]).round(2)
```

Data Augmentation	*Not applicable*

Denoising	
Edge Detection	*Not applicable*

```
## Lower Casing
df["reviews_list"] = df["reviews_list"].str.lower()
## Removal of Puctuations
import string
PUNCT_TO_REMOVE = string.punctuation
def remove_punctuation(text):
  """custom function to remove the punctuation"""
  return text.translate(str.maketrans('', '', PUNCT_TO_REMOVE))
df["reviews_list"] = df["reviews_list"].apply(lambda text: remove_punctuation (text))
```

Color Space Conversion	*Not applicable*

Image Cropping	*Not applicable*
Batch Normalization	*Not applicable*